

GRAND FORKS AFB, ND, USA

WMO: 727575

Lat: 47.967N

Lon: 97.400W

Elev: 913

StdP: 14.22

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94925

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.1	-15.3	-25.7	1.4	-19.1	-21.0	1.8	-14.5	32.1	9.5	28.5	10.8	7.8	310	0.678

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.9	89.1	72.1	86.0	70.4	83.1	68.5	76.4	83.7	73.7	81.8	71.5	79.6	13.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.0	131.0	80.0	71.7	121.2	78.5	68.7	109.3	75.6	40.5	84.4	37.9	80.7	35.9	78.5	88.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.9	24.9	21.8	DB	-26.3	94.9	6.8	2.4	-31.2	96.6	-35.1	98.0	-38.9	99.3	-43.9	101.0	(4)
(5)			WB	-25.7	80.3	6.5	3.3	-30.4	82.7	-34.2	84.6	-37.9	86.5	-42.6	88.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.7	7.1	11.7	25.3	40.5	54.5	64.9	69.5	67.7	58.7	44.3	27.9	14.4		
	DBStd	24.27	14.53	13.96	14.04	10.39	8.71	6.82	5.40	5.91	8.46	9.22	11.58	13.01		
	HDD50	5547	1329	1072	767	311	54	1	0	0	19	227	664	1103		
	HDD65	9281	1794	1492	1229	735	341	83	20	39	224	643	1113	1568		
	CDD50	2153	0	0	3	26	192	447	606	549	279	50	1	0		
	CDD65	414	0	0	0	0	15	79	160	123	35	2	0	0		
	CDH74	4602	0	0	0	17	258	881	1620	1347	441	38	0	0		
	CDH80	1395	0	0	0	3	65	258	488	442	132	7	0	0		
	WSAvg	10.5	11.4	11.5	11.9	11.5	11.2	9.5	8.4	8.6	9.8	10.9	11.0	11.0		
	PrecAvg	21.3	0.4	0.4	0.7	1.1	3.2	3.8	3.7	2.9	2.0	1.8	0.9	0.6		
(15) Precipitation	PrecMax	25.7	1.1	1.4	2.3	2.6	9.0	7.9	7.5	5.6	5.4	6.0	4.0	1.3		
	PrecMin	14.2	0.0	0.0	0.0	0.2	0.9	1.0	1.0	0.7	0.3	0.0	0.0	0.0		
	PrecStd	3.1	0.4	0.4	0.7	0.6	1.8	1.6	1.8	1.1	1.2	1.4	0.9	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.6	45.9	63.8	75.3	87.0	92.7	91.2	93.3	89.5	79.1	61.1	44.5		
		MCWB	37.3	39.9	53.1	55.6	65.0	71.9	75.0	75.1	71.1	61.3	51.7	38.2		
	2%	DB	37.4	39.4	55.1	69.7	81.0	86.0	88.0	88.5	83.9	70.4	53.9	39.0		
		MCWB	34.2	36.6	48.0	53.4	61.5	68.8	74.0	71.4	66.7	56.7	45.3	35.2		
	5%	DB	32.6	36.1	48.3	63.6	76.0	82.3	85.3	84.8	79.3	64.1	48.3	35.6		
		MCWB	31.1	34.3	43.2	50.0	58.6	66.6	72.3	69.9	63.8	53.5	42.6	33.2		
	10%	DB	27.9	32.3	42.6	58.6	71.8	79.1	82.3	81.5	74.6	59.3	43.3	31.8		
		MCWB	26.6	31.0	39.4	47.6	56.9	65.2	70.0	67.8	62.2	51.2	39.0	30.2		
	0.4%	WB	38.1	41.5	56.3	57.6	68.4	77.3	81.0	80.1	73.8	63.8	52.1	39.1		
		MCDB	41.3	44.3	60.7	69.0	77.9	84.3	84.3	85.8	83.5	73.1	59.5	43.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	34.6	37.1	48.7	54.1	65.4	73.1	77.3	75.0	69.8	59.3	46.7	36.0		
		MCDB	37.2	39.5	54.8	66.2	75.7	81.5	84.0	82.8	78.6	67.5	52.3	38.7		
	5%	WB	31.3	34.3	43.2	51.3	62.3	69.9	74.7	72.4	66.8	55.3	43.0	33.0		
		MCDB	33.1	35.9	47.8	61.9	70.1	77.6	81.9	81.0	75.1	63.0	48.0	35.1		
	10%	WB	26.8	30.9	39.1	48.4	59.4	67.6	72.4	70.2	64.1	51.5	39.1	30.0		
		MCDB	27.9	32.4	42.6	57.7	67.9	75.3	79.9	77.9	71.9	58.5	43.0	31.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.8	17.5	17.4	21.4	23.6	21.6	21.9	23.7	23.0	19.5	16.7	15.3		
		MCDBR	18.9	18.4	25.1	31.7	30.5	26.7	24.9	27.8	29.6	28.9	24.9	19.1		
		MCWBR	16.8	15.2	17.5	17.4	15.4	13.1	13.5	13.2	14.8	16.6	16.9	15.8		
		MCDBR	19.0	17.0	24.0	29.7	24.0	22.4	22.4	24.2	24.8	25.6	23.7	18.2		
	5% WB	MCWBR	17.1	14.1	17.7	17.3	14.5	13.5	14.1	13.6	14.7	16.2	16.6	15.2		
(40) Clear-Sky Solar Irradiance	taub		0.283	0.300	0.300	0.353	0.368	0.381	0.381	0.383	0.348	0.305	0.275	0.261		
	taud		2.287	2.268	2.445	2.353	2.373	2.366	2.371	2.366	2.455	2.546	2.517	2.424		
	Ebn at Noon		249	272	294	285	284	279	278	272	273	271	258	249		
	Edh at Noon		23	30	30	36	37	37	37	36	30	23	19	18		
(44) All-Sky Solar Radiation	RadAvg		500	883	1241	1545	1742	1899	2012	1678	1231	762	450	345		
	RadStd		53	78	128	141	112	141	87	115	87	110	54	40		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page